(12) Federated Learning and Privacy-Preserving Rags

(V1) Introduction -

→ Federated Learning →

→ Federated Learning allows multiple devices or organisations to train AI models collaboratively sharing the data

→ Privacy-Preserving →

→ Privacy-Preserving rags enhances retrieval augmental generation systems by incorporating techniques like encryption and differential privacy to protect sensitive information during retrieval and model training.

→ Prerequisites → Experience with AI/ML systems, Rag.

(V2) Federated Learning for Retrieval -

→ Federated Learning keeps your information safe by only sharing model updates, not your personal data with a central server.

→ Advantages of Federated Learning →

① Data Privacy → keep raw data on local devices transmitting only model updates

② Reduced Bandwidth usage → Lowest network bandwidth requirements and cost by model update transfer instead of full data sets.

③ Model Generalization → Aggregates insights from diverse data sources, improving models ability to improvise

④ Regulatory Compliance → Ensure adherence of data protection law like GDPR and HIAA by preventing cross border and intra entity data transfer

⑤ Scalability → efficiently scales across distributed devices, managing heterogeneous data and varying computational capacities without central data storage.

V3 → Federated Learning for Rag →

→ Federated Learning enhances rag using fed techniques that allows decentralised collaboration while keeping personal data private.

→ Enhancing genAI →

① Combining Techniques in AI →

② Enhanced Content Generation

③ Addressing challenges with Advanced Techniques.

→ FedAvg → stands for Federated Average →

- Aggregates model updates
- Relieves only ~~updat~~ results
- Update in a global model
- Best for privacy-focused on training across multiple decentralised devices

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(V4) Privacy-Preserving Techniques in Federated Learning →

→ Key preserving Techniques to help data secure while enabling collaborative learning model.

→ Data Privacy in AI →

① Decentralized Training → Federated learning enables decentralised model training by allowing multiple device or organisations to collaborate without sharing their ~~[illegible]~~ data.
→ This approach improves data privacy, but ensuring that sensitive information is never compromised and remains critical

② Privacy Enhancement → For privacy enhancement we can use several privacy preserving techniques that are integrated into federated learning system to protect data during the training process.

③ Privacy-Preserving Techniques → We have privacy techniques, ~~some of them are~~ most commonly used methods are-
(a) homomorphic encryption
(b) Differential privacy
(c) secure multi-party computation (SMPC)

ⓐ homomorphic encryption →
→ In homomorphic encryption →
→ You encrypt your data (lock the box of apples)
→ someone else (like a cloud server / vendor) performs calculations (adding apples) without decrypting it (counting apples in the box)
→ you decrypt the result later.

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→ homomorphic method is useful since the data stays secret the whole time, even during processing

→ In one line →
homomorphic encryption lets you do maths on encrypted data without even decrypting it.

② Differential Privacy →

→ It is a method that hides your individual data by mixing it with randomness, while still allowing useful patterns to be learned from the overall dataset.

→ so for example →
if someone tries to guess your data →
- They can't be sure if you were in the dataset or not
- your privacy stays protected.

→ Differential privacy helps organisations share or analyze data without exposing individuals.

→ used by
1. Apple (to improve typing suggestions)
2. Google (for user analytics)
3. US Census (to protect citizen data)
4. Meta, Amazon, Microsoft (for large-scale privacy)

→ in one line →
Differential privacy adds controlled randomness so no one can trace results back to a specific person.

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